MAHMOUD SALAMA

SOFTWARE ENGINEERING & APPLICATIONS LEADER | CLOUD-NATIVE MODERNIZATION

Engineering Leadership | C#/.NET & APIs | Cloud-Native Modernization | SDLC & Quality | DevOps & CI/CD | Production Readiness

+20 122 015 6077 | ma7moud.salamaaa@gmail.com | Cairo, Egypt

linkedin.com/in/mahmoud-salama-30249b34 | mahmoud-salama.vercel.app | github.com/MahmoudSalamaaa

PROFESSIONAL SUMMARY

Software engineering and applications leader with 18+ years across Egypt and Oman, progressing from hands-on Linux, C#/.NET and full-stack engineering through development-team leadership to application-platform, architecture and national-scale engineering leadership across 200+ technology projects across the career. Leads software and application capabilities across a multidisciplinary technology organization, combining deep technical judgment with SDLC, QA/UAT, release, integration and production-readiness governance across C#/.NET, web platforms, APIs, databases, reporting and production operations.

CORE SKILLS

Software Engineering Leadership | C#/.NET, APIs & Full-Stack Engineering | Application Development & Cloud-Native Modernization | Engineering Operating Model | SDLC & Release Governance | Architecture & Design Reviews | QA/UAT & Production Readiness | API & Integration Engineering | Application Security & Resilience | Technical Debt & Maintainability | DevOps & CI/CD Governance | Cross-Functional Engineering Leadership

SELECTED ENGINEERING IMPACT

- Led MedIQ re-architecture and replacement through UPA's in-house technology organization; the national procurement, supply and inventory platform won the Pharmaconex Awards 2026 – Excellence in Digital Transformation.
- Directed the completed Microsoft Dynamics 365 enterprise integration across procurement, supply, inventory, finance and reporting, governing implementation quality, integration contracts, testing, security, change control and production transition.
- Own application-engineering governance across MedIQ and related platforms, covering architecture/design decisions, SDLC, QA/UAT, release readiness, performance testing, production support and continuous improvement.
- **Lead application engineering and platform delivery for MedIQ and connected national-scale systems serving 38,000+ current users across 5,900+ government healthcare entities and nearly 90,000 affiliated operational locations.**
- **Built engineering delivery depth across major Omani government platforms and 150+ major and mid-sized engagements, progressing from hands-on full-stack development to leading development teams.**
- Created reusable CMS, reporting and shared application patterns for multilingual and RTL solutions, standardizing delivery across client and government projects.

SYSTEMS ANALYSIS & DELIVERY

Translate stakeholder needs into AS-IS/TO-BE workflows, functional requirements, user stories, acceptance criteria and interface/data mappings; use UML, Visio and Jira to align engineering, business validation and change control.

Govern scope, milestones, RAID/dependencies, capacity, QA/UAT, cutover, hypercare and operational handover; distinguish technical acceptance from contractual approval.

PROFESSIONAL EXPERIENCE

Egyptian Authority for Unified Procurement (UPA) **05/2025 - Present**

Head of the Central Administration for Information Systems & Digital Transformation **Cairo, Egypt**

- **Lead engineering and application-platform direction across MedIQ and connected services within a 54-person organization with 9 direct reports; led technical delivery of the greenfield, bi-directional SAP Strategic Medical Warehouses integration across interfaces, data mapping, security, UAT and production support.**
- **Designed and developed production healthcare REST APIs integrating UPA with external health entities, implementing token-based authentication, structured JSON contracts, beneficiary/prescription/dispensing retrieval, GTIN-enabled product identification and standardized error handling.**
- Direct architecture and design decisions with relevant technical teams; use PoCs when needed, assess architecture risk, manage technical debt and balance maintainability, scalability, security, cost and delivery constraints.
- Govern production architecture across ReactJS/Node.js, C#/.NET APIs, PostgreSQL, Redis, containers and Private Cloud; designed reusable reporting and practical ODS patterns, with data migration and reconciliation across procurement and invoice records.
- Drive technical delivery for MedIQ E-Tender, the completed Dynamics 365 enterprise integration and the HTA system/platform; directly designed/developed the HTA technology solution while ensuring implementation quality, testability, acceptance, production readiness and supportability.
- Own engineering production-readiness controls including capacity planning, load/performance testing, QA/release gates, monitoring/logging, incident escalation and RCA, backup/recovery and DR coordination.

Egyptian Authority for Unified Procurement (UPA) **10/2020 - 05/2025**

General Manager - Systems, Applications & Technical Support **Cairo, Egypt**

- Directed application modernization from established .NET Framework/Telerik solutions toward ReactJS/Node.js services, C#/.NET APIs, PostgreSQL, Redis, Docker and Apache Superset while preserving operational continuity.

- Strengthened SDLC execution through iterative planning, engineering standards, architecture/design and code reviews, QA/UAT and release gates, documentation, incident escalation, root-cause analysis and knowledge transfer.
- Expanded engineering capability as workload grew, strengthening QA/testing, DevOps, mobile and application functions while tightening the path from requirements and architecture through engineering, deployment and production support.
- Converted manual, paper- and spreadsheet-heavy processes into governed digital workflows with role-based access, approvals, validation, audit trails, reporting and defined operational ownership.

Integral Solutions LLC **2016 - 2020**

Development Team Leader & Senior Full-Stack Developer **Muscat, Oman**

- Led a 20-person development team across 15+ Omani government and enterprise platforms while remaining hands-on in solution design, full-stack development, testing, deployment, releases and production support.
- **Contributed extensively to Oman's National Educational Portal (~1M users), delivering LMS capabilities across courses/content, exams, attendance, student/teacher accounts, RBAC, reporting and integrations; also contributed to Maktabi and the Ministry of Education Recruitment System. Developed Maktabi correspondence workflows, reports and database components, with functional testing, release validation and production support.**
- Designed and delivered secure government workflows, reporting, database and integration components across C#/.NET, ASP.NET Web Forms, SQL Server and MySQL.

Keyframe Egypt **04/2011 - 05/2016**

Development Team Leader & Senior Full-Stack Developer **Cairo, Egypt**

- **Delivered or led 150+ major and mid-sized engagements across government, healthcare, education and commercial sectors, progressing from hands-on full-stack engineering to development-team leadership.**
- Owned analysis, C#/.NET engineering, database/reporting design, testing, deployment and support across multilingual and RTL solutions while balancing concurrent client priorities.
- Delivered Yashfeen, the healthcare solution associated with confirmed ITIDA competition/award recognition, and created reusable CMS, reporting and shared application patterns across multiple projects.

Egyptian Armed Forces **04/2008 - 10/2010**

Linux Systems & Web Developer **Egypt**

- Built hands-on systems depth through Linux From Scratch, kernel modules, optimization/hardening and secure ASP.NET/SQL Server application development.

ENGINEERING LEADERSHIP & TECHNICAL TOOLKIT

Engineering Leadership: Software engineering management, application development, multidisciplinary team leadership, estimation and capacity planning, architecture/design and code reviews, SDLC governance, QA/UAT, release management, production support, maintainability, technical debt and continuous improvement.

Modern Engineering: C#/.NET, ASP.NET, ReactJS, Node.js, REST/GraphQL APIs, PostgreSQL, SQL Server, MySQL, Redis, Docker; Kubernetes (working knowledge), Git/TFS, Jenkins, CI/CD governance, Postman, Swagger/OpenAPI.

Full-Stack Foundation: ASP.NET Web Forms, .NET Framework, Entity Framework, LINQ, JavaScript, jQuery, AJAX, HTML/CSS, Bootstrap, REST/SOAP, Telerik, Kendo, Infragistics, Crystal Reports, Umbraco and Kentico.

SELECTED ENGINEERING & GOVERNANCE CONTRIBUTIONS

- National Pharmaceutical Track & Trace Program (2021-2024) - contributed to multi-agency governance, compliance and interoperability.
- Board Member - National Blood Operations Oversight (2021-2024), representing UPA and contributing to digital governance, data integrity and technology standards.
- Member - Egypt Digital Transformation Committee supporting the World Bank Group B-READY Assessment (2025-2026).

EDUCATION & LANGUAGES

B.Sc. in Computer Science, Faculty of Computers & Informatics, Zagazig University | 2003-2007

Arabic - Native | English - Professional Working Proficiency